Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF COMPUTER ENGINEERING AND INFORMATION SYSTEM

DEPARTMENT: SOFTWARE ENGINEERING

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REG NO: 23MIS0587

COURSE NAME: Agile Development Process and Devops Lab

COURSE CODE: ISWE406P

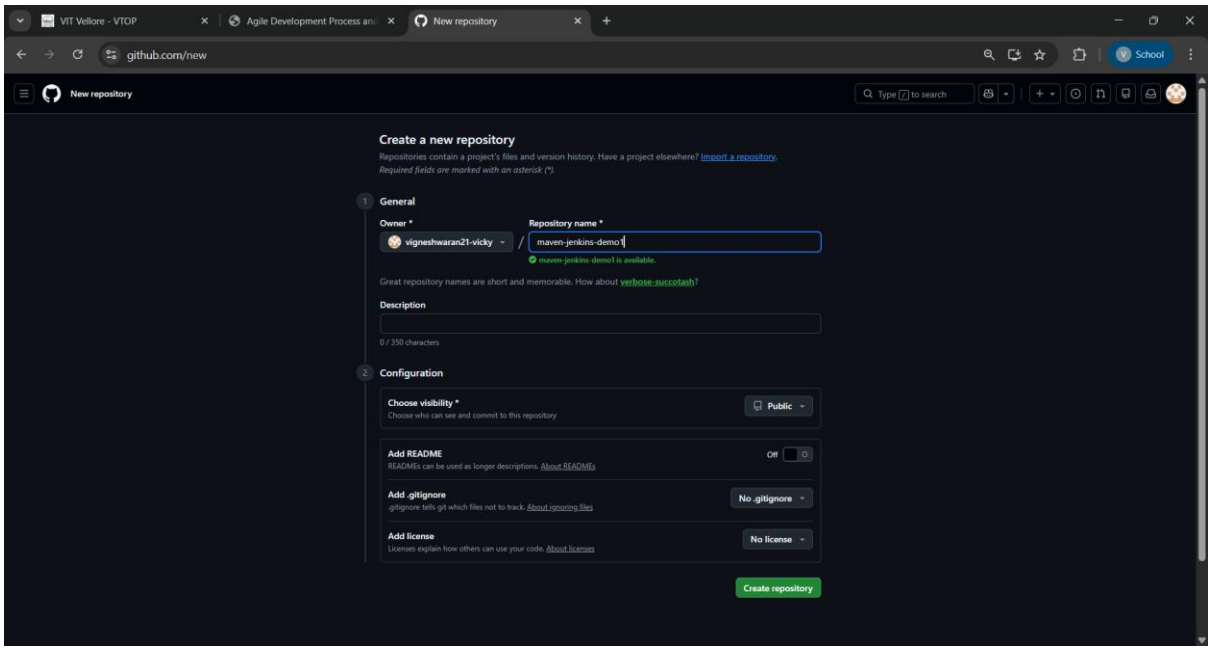
SLOT: L3 + L4

DIGITAL ASSIGNMENT –III

(**Maven**)

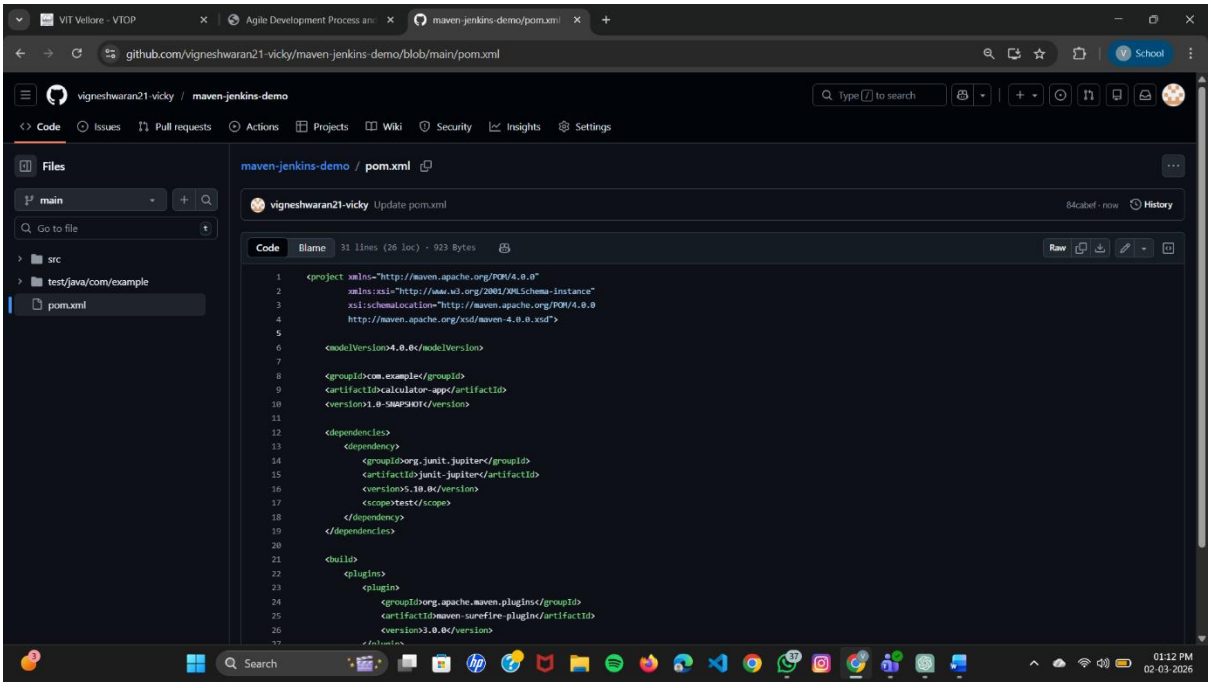
1.Create a Maven Project in Git

• Create a new Git repository

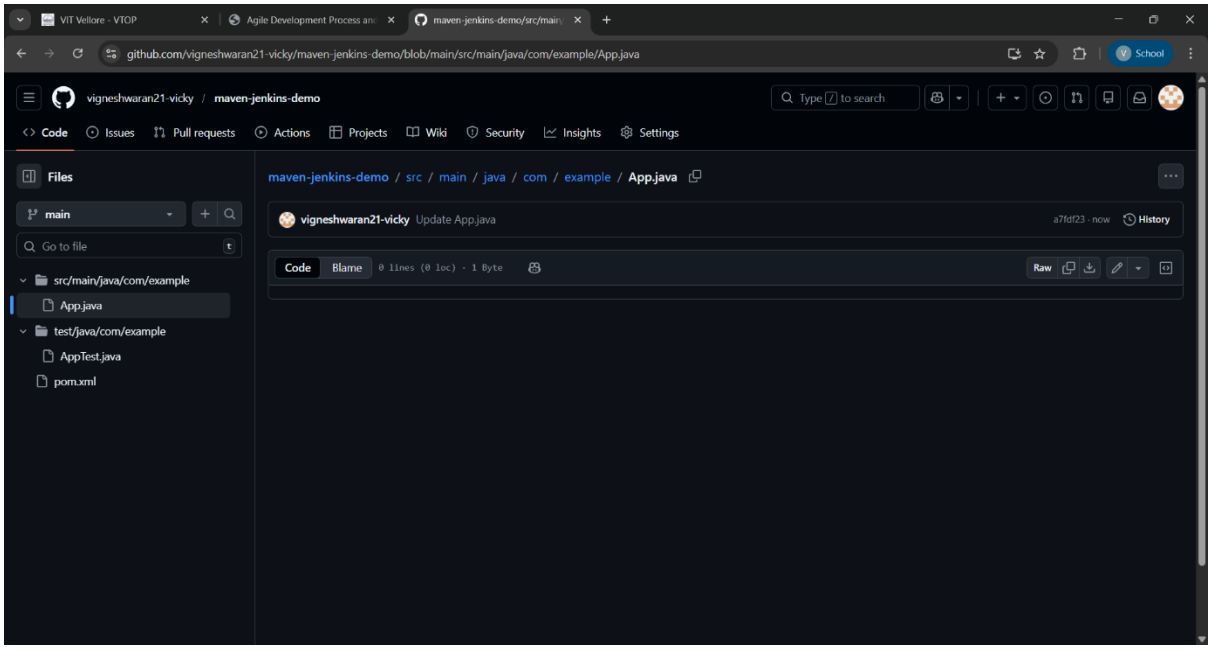


• Create a simple Maven project structure manually, including the following files and folders:

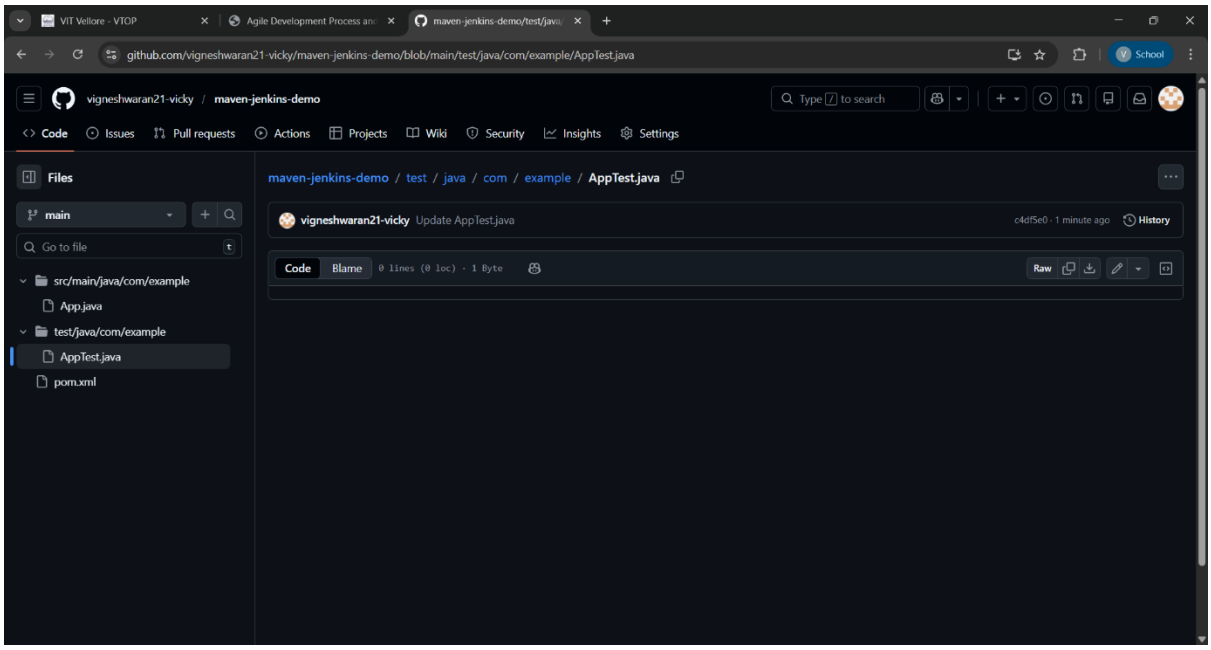
❖ pom.xml



❖ src/main/java/com/example/App.java

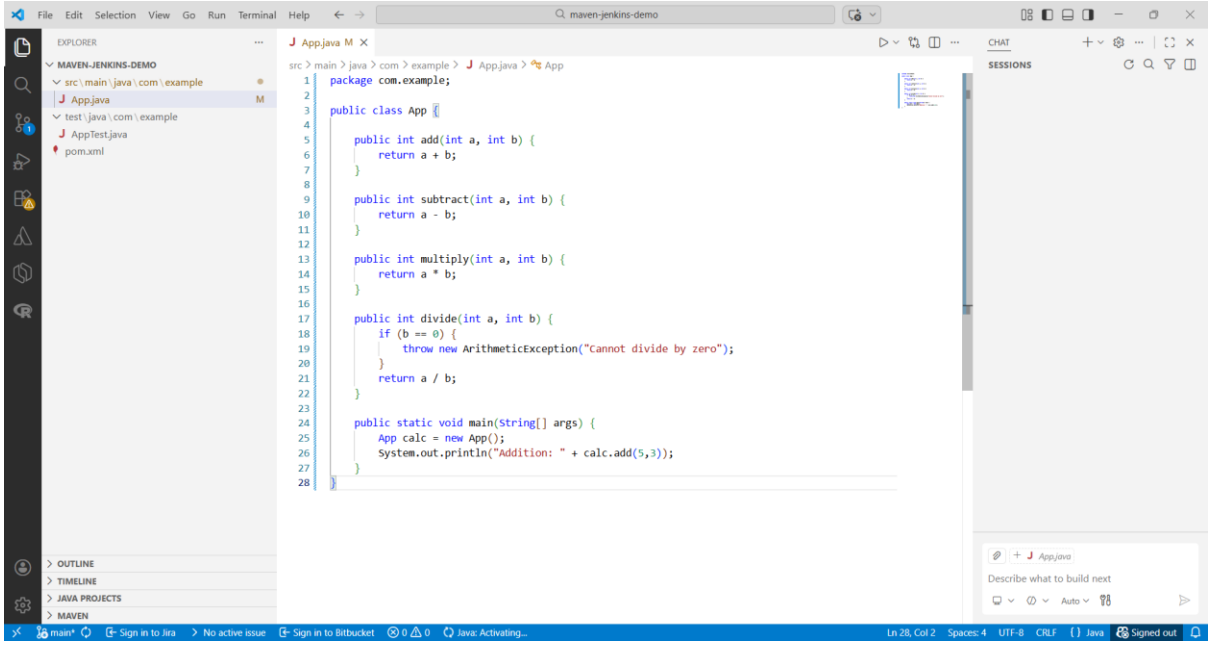


❖ src/test/java/com/example/AppTest.java

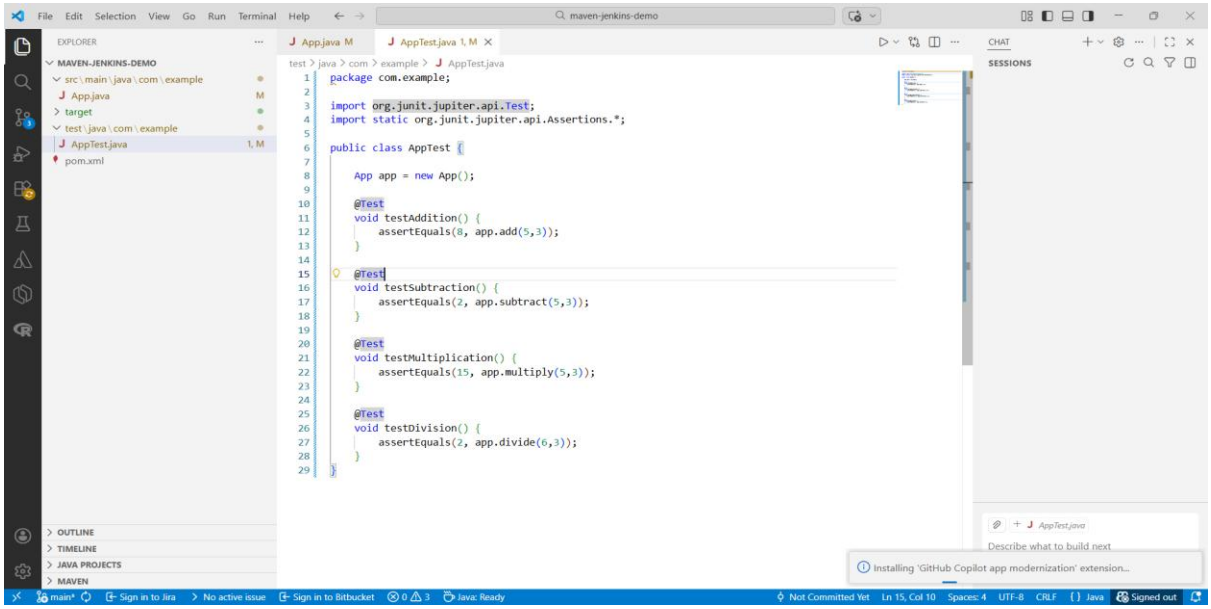


2. Implement the Java Program

- Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios:
  - o Student Grade Calculator
  - o Bank Account Management System
  - o Employee Salary Calculator
  - o Library Book Management System
  - o Simple Calculator Application

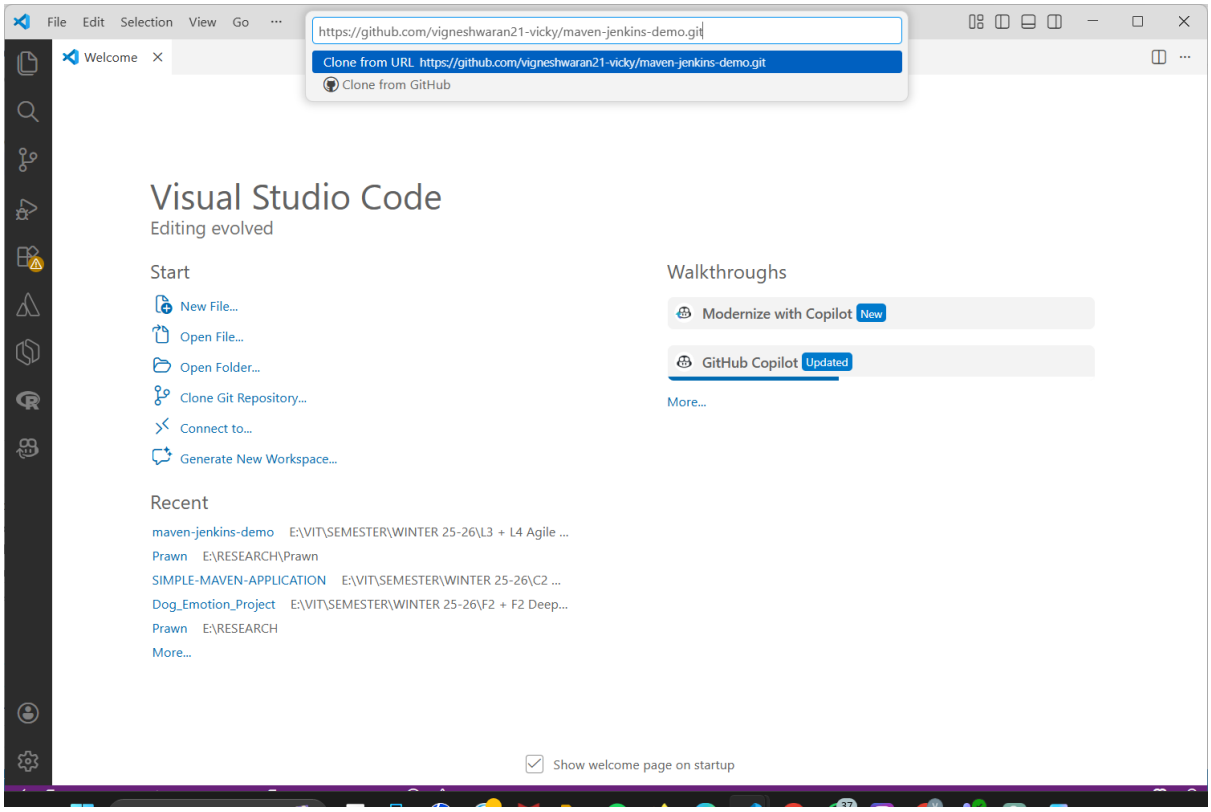


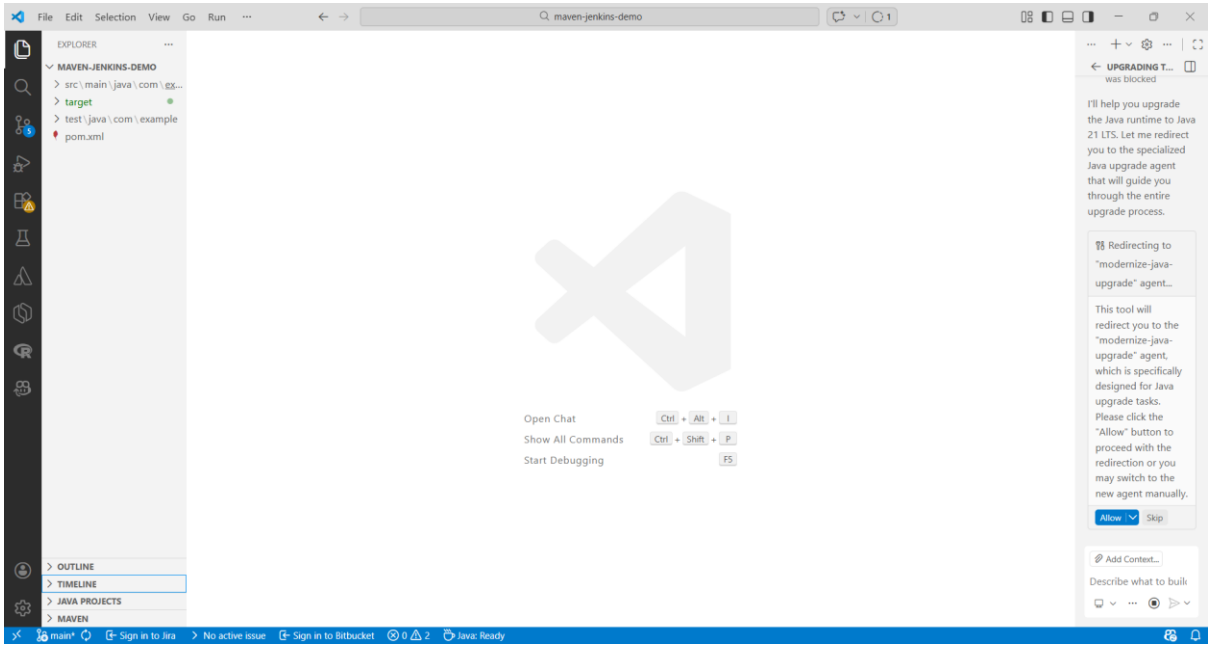
- Write a corresponding unit test in AppTest.java.

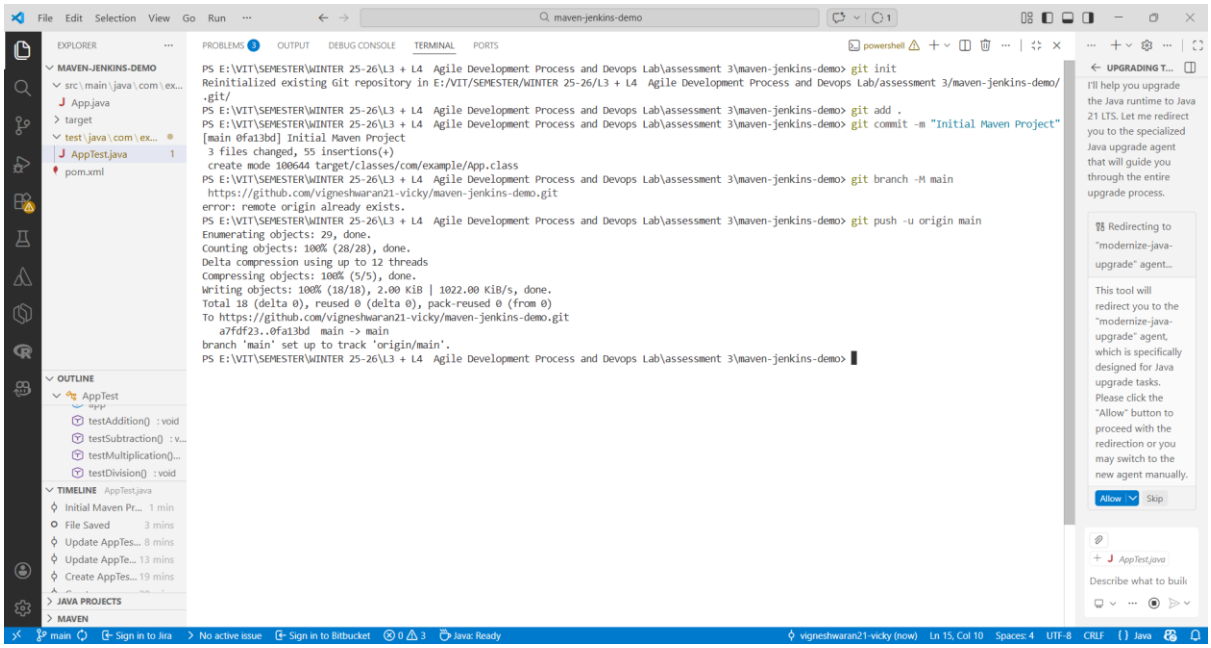


3. Push and Clone the Repository

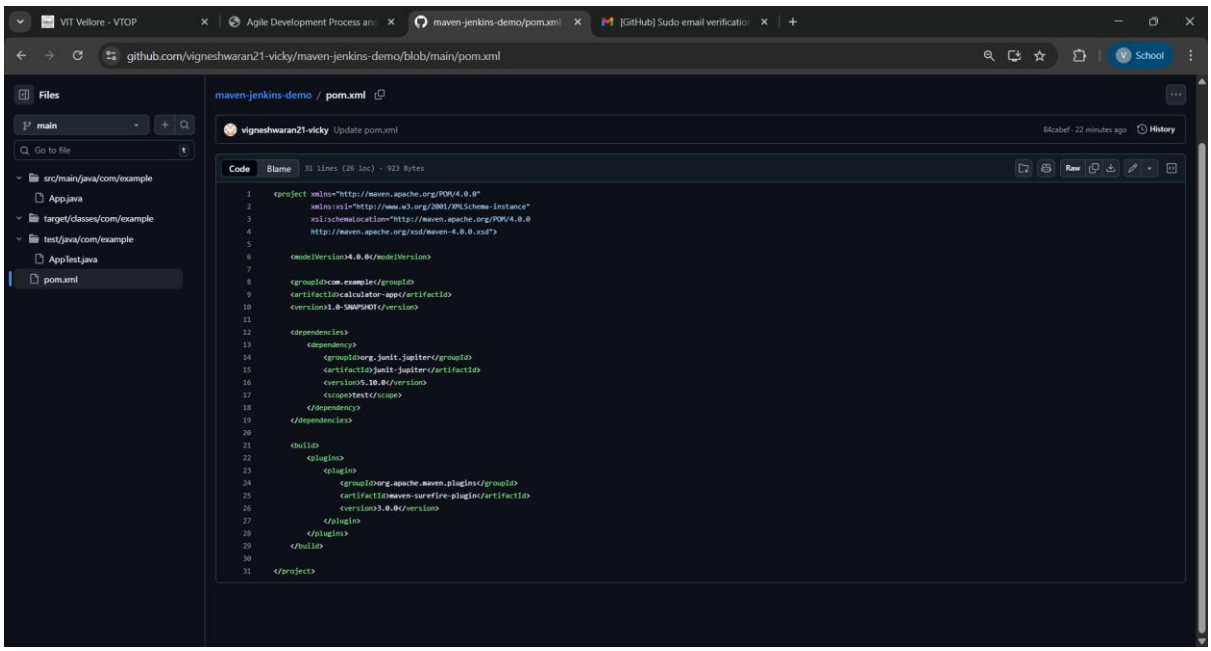
- **Clone the repository using Visual Studio Code.**







- **Verify the project structure.**



github.com/vigneshwaran21-vicky/maven-jenkins-demo/blob/main/src/main/java/com/example/App.java

Files

main

Go to file

- src/main/java/com/example
  - App.java
- target/classes/com/example
- test/java/com/example
  - AppTest.java
- pom.xml

maven-jenkins-demo / src / main / java / com / example / App.java

vigneshwaran21-vicky Initial Maven Project 0fa13bd · 4 minutes ago History

Code Blame 28 lines (22 loc) · 551 Bytes Raw

```java
package com.example;

public class App {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Cannot divide by zero");
        }
        return a / b;
    }

    public static void main(String[] args) {
        App calc = new App();
        System.out.println("Addition: " + calc.add(5,3));
    }
}
```

github.com/vigneshwaran21-vicky/maven-jenkins-demo/blob/main/test/java/com/example/AppTest.java

Files

main

Go to file

- src
- target/classes/com/example
- test/java/com/example
  - AppTest.java
- pom.xml

maven-jenkins-demo / test / java / com / example / AppTest.java

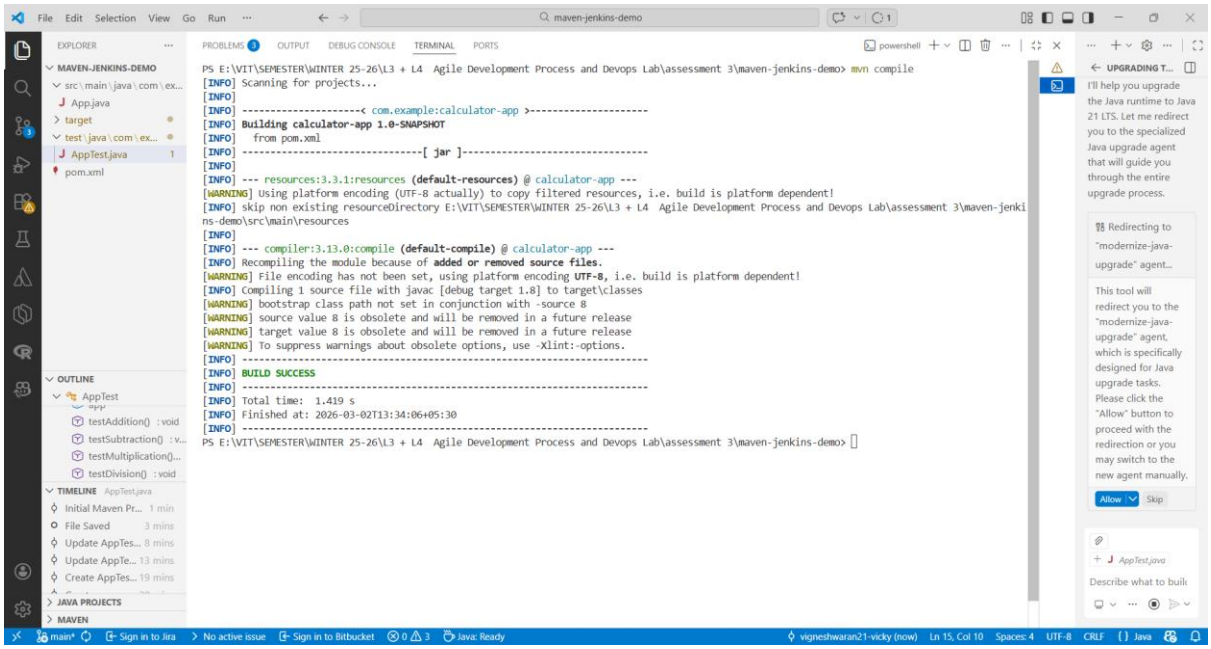
vigneshwaran21-vicky Initial Maven Project 0fa13bd · 2 minutes ago History

Code Blame 29 lines (22 loc) · 589 Bytes Raw

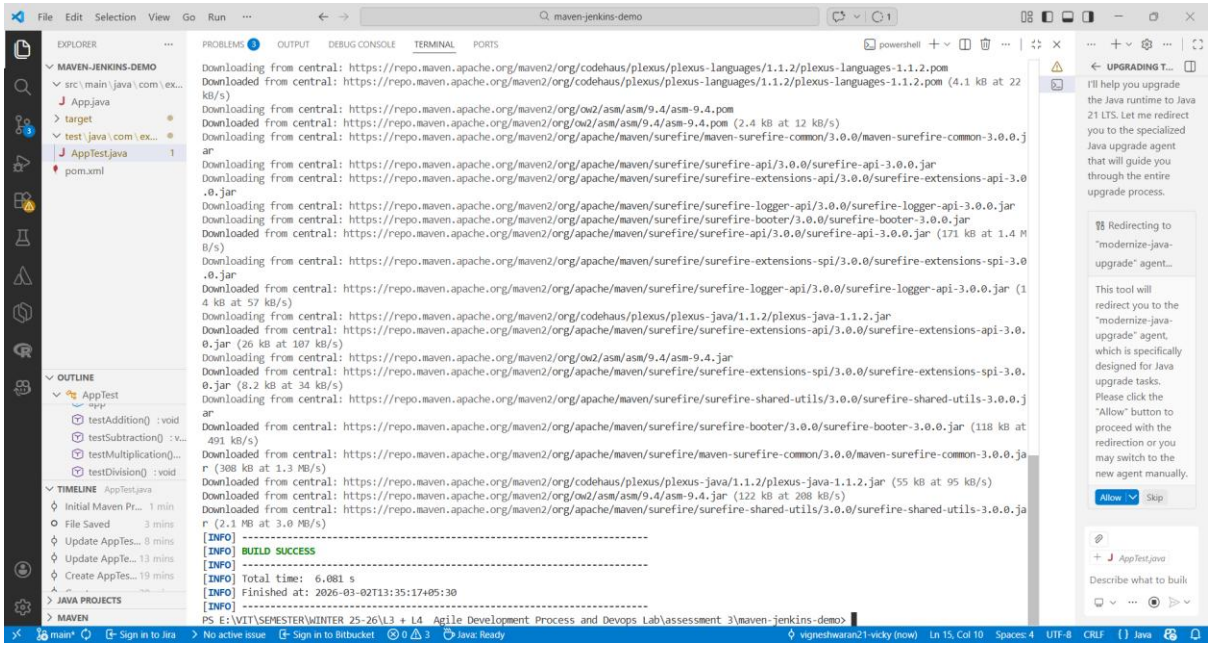
```java
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class AppTest {

    App app = new App();

    @Test
    void testAddition() {
        assertEquals(8, app.add(5,3));
    }

    @Test
    void testSubtraction() {
        assertEquals(2, app.subtract(5,3));
    }

    @Test
    void testMultiplication() {
        assertEquals(15, app.multiply(5,3));
    }

    @Test
    void testDivision() {
        assertEquals(2, app.divide(6,3));
    }
}
```

4. Build the Maven Project

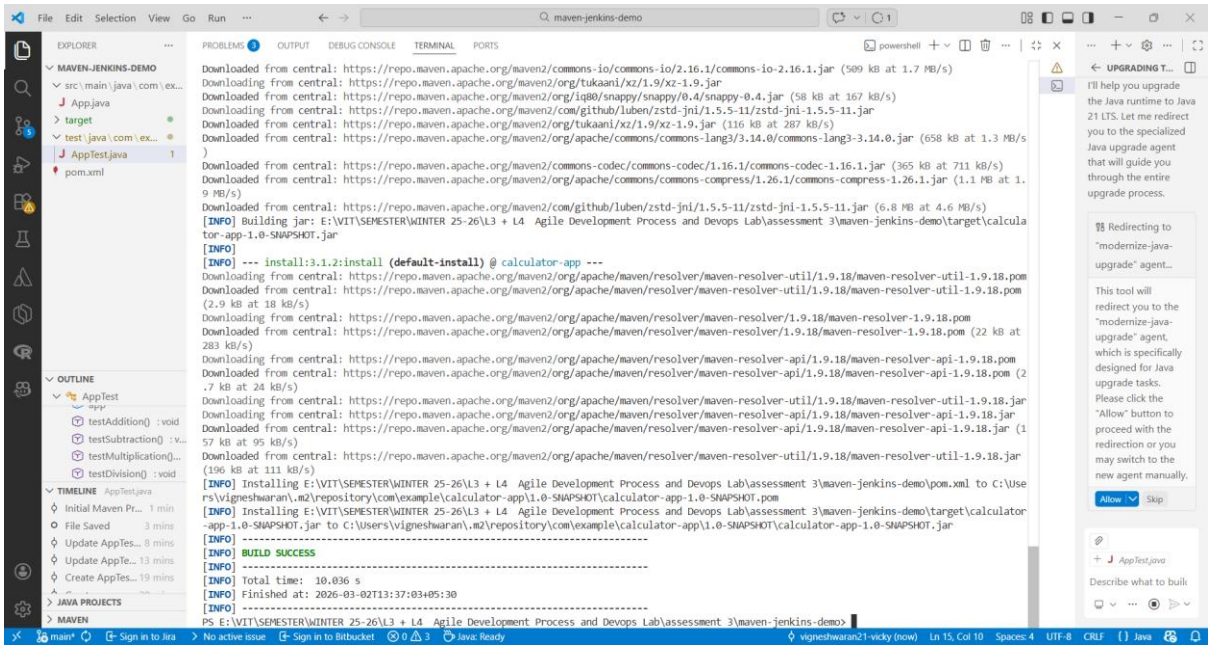
- **Use Maven commands in Visual Studio Code terminal to:**
  - Compile the project



  - Run tests

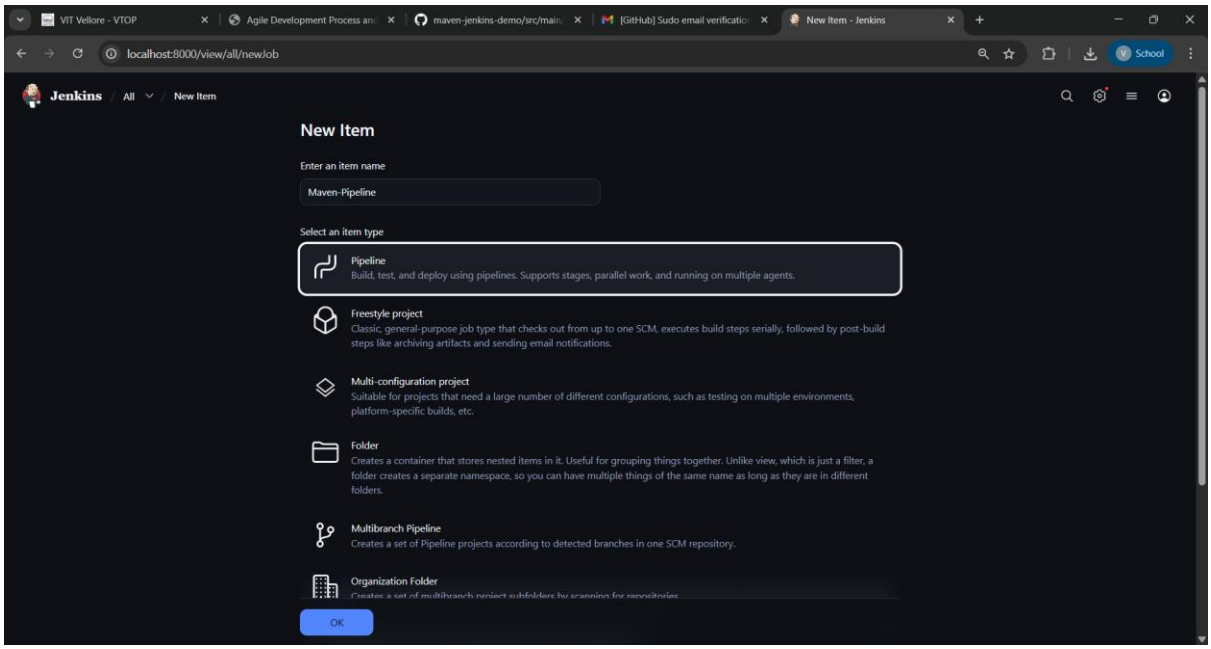


- Build the project (mvn clean install)

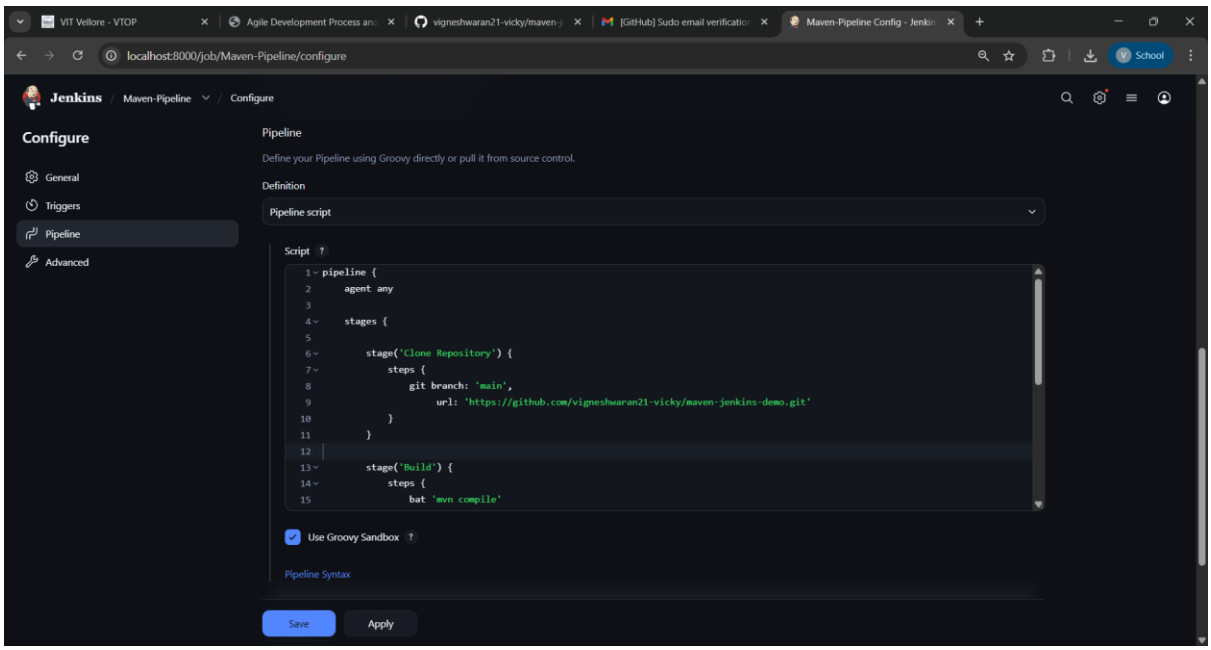


5. Create and Run Jenkins Pipeline

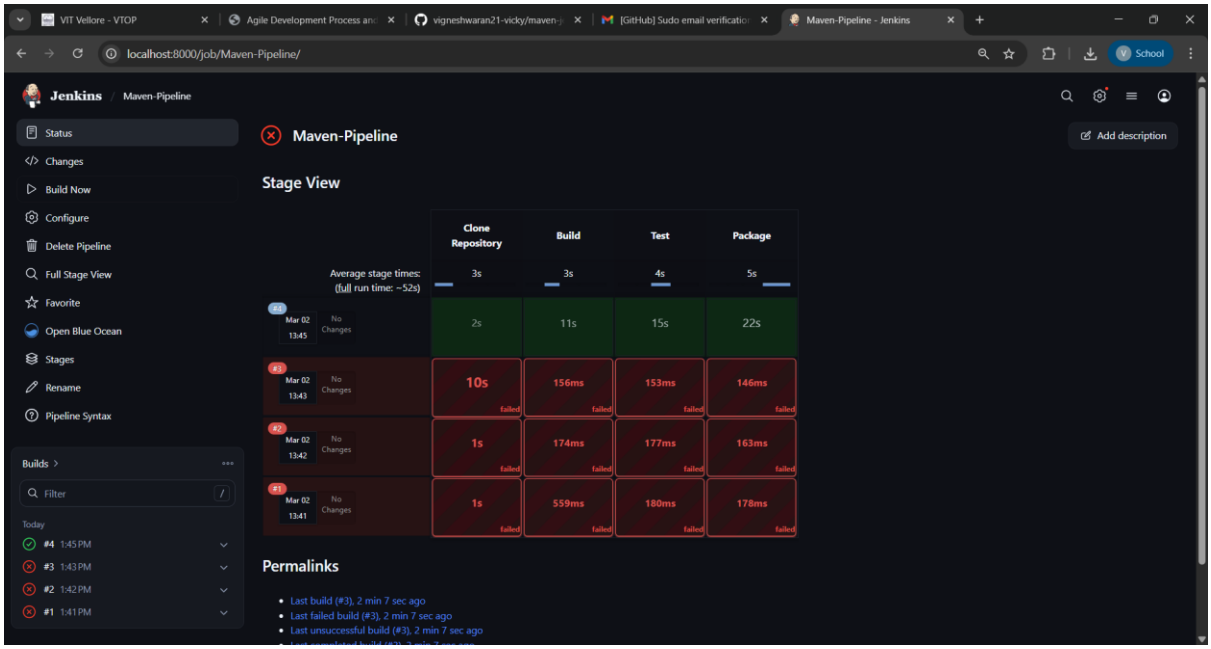
• Create a new Pipeline project in Jenkins.

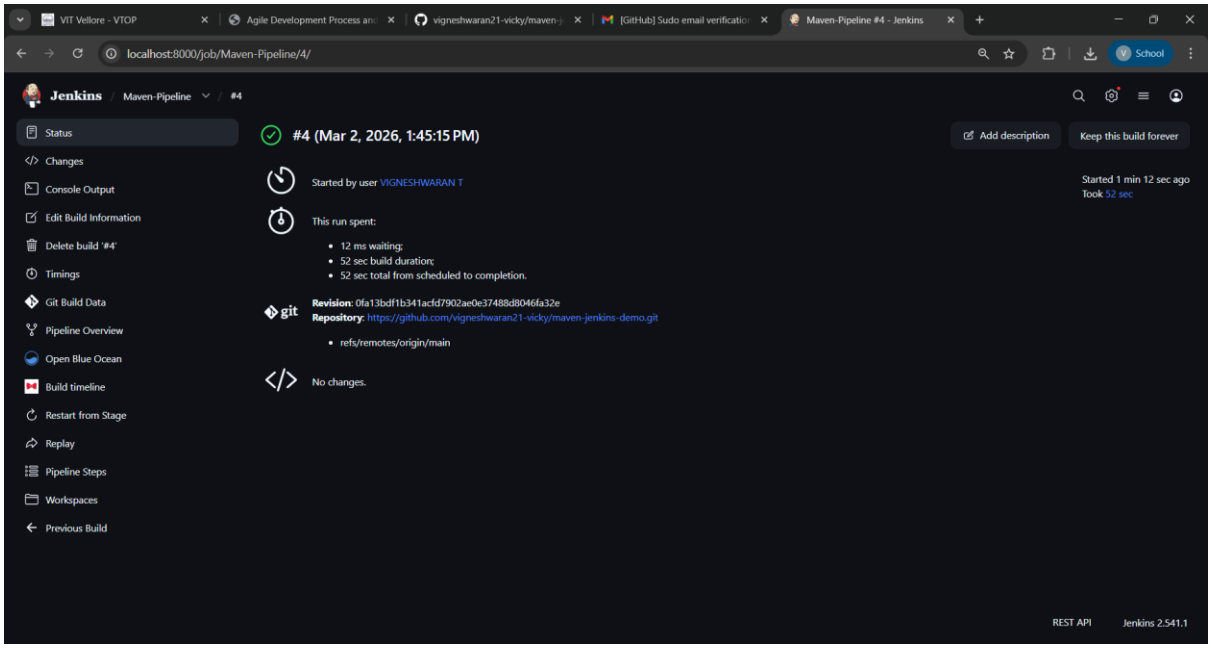


- Create a pipeline script



- Execute the pipeline and verify successful build completion.

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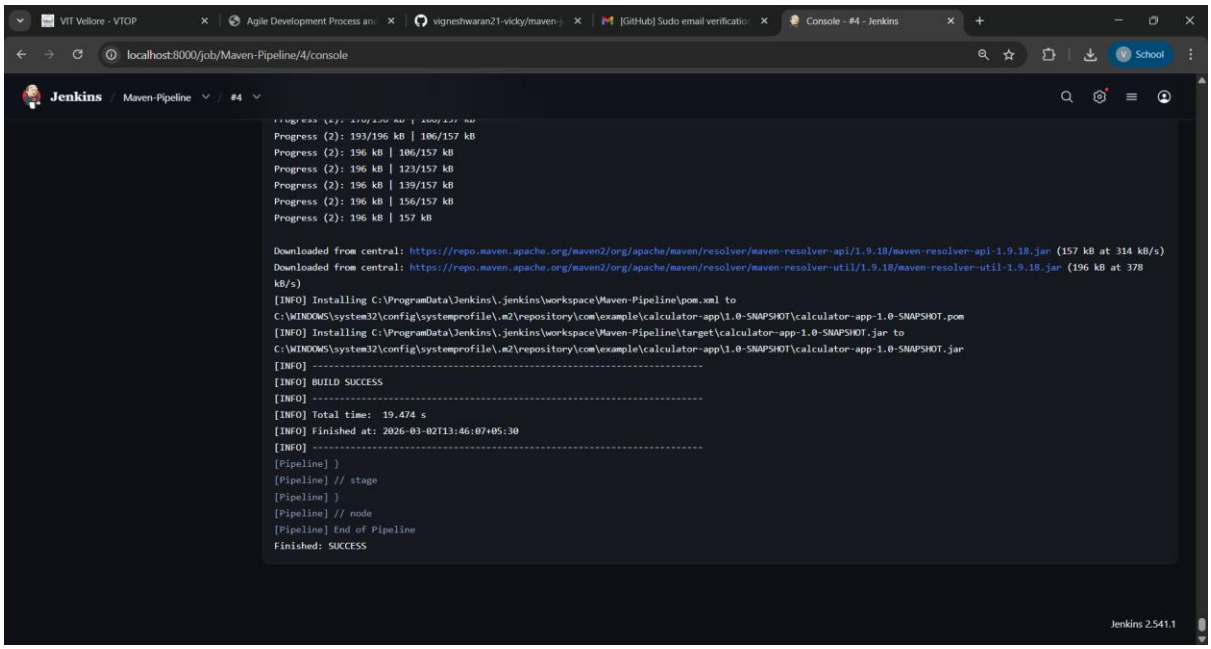
Maven-Pipeline #4 - Jenkins
localhost:8000/job/Maven-Pipeline/4/
School
Jenkins / Maven-Pipeline / #4
Status
Changes
Console Output
Edit Build Information
Delete build '#4'
Timings
Git Build Data
Pipeline Overview
Open Blue Ocean
Build timeline
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#4 (Mar 2, 2026, 1:45:15 PM)
Add description
Keep this build forever
Started by user VIGNESHWARAN T
Started 1 min 12 sec ago
Took 52 sec
This run spent:
12 ms waiting;
52 sec build duration;
52 sec total from scheduled to completion.
Revision: 0fa13bdf1b341acfd7902ae0e37488d8046fa32e
Repository: https://github.com/vigneshwaran21-vicky/maven-jenkins-demo.git
refs/remotes/origin/main
No changes.
REST API
Jenkins 2.541.1


VIT Vellore - VTOP
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Console - #4 - Jenkins
localhost:8000/job/Maven-Pipeline/4/console
School
Jenkins / Maven-Pipeline / #4
Progress (2): 193/196 kB | 106/157 kB
Progress (2): 196 kB | 106/157 kB
Progress (2): 196 kB | 123/157 kB
Progress (2): 196 kB | 139/157 kB
Progress (2): 196 kB | 156/157 kB
Progress (2): 196 kB | 157 kB
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-api/1.9.18/maven-resolver-api-1.9.18.jar (157 kB at 314 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/resolver/maven-resolver-util/1.9.18/maven-resolver-util-1.9.18.jar (196 kB at 378 kB/s)
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Maven-Pipeline\pom.xml to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\calculator-app\1.0-SNAPSHOT\calculator-app-1.0-SNAPSHOT.pom
[INFO] Installing C:\ProgramData\Jenkins\.jenkins\workspace\Maven-Pipeline\target\calculator-app-1.0-SNAPSHOT.jar to C:\WINDOWS\system32\config\systemprofile\.m2\repository\com\example\calculator-app\1.0-SNAPSHOT\calculator-app-1.0-SNAPSHOT.jar
[INFO] --
[INFO] BUILD SUCCESS
[INFO] --
[INFO] Total time: 19.474 s
[INFO] Finished at: 2026-03-02T13:46:07+05:30
[INFO] --
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
Jenkins 2.541.1